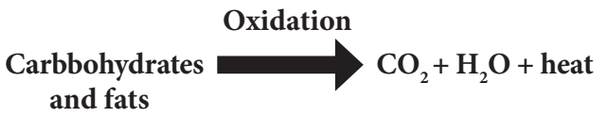
9.7 Energy

Energy is the capacity to do work. Energy must be supplied regularly to meet the needs of the body for survival. The body needs energy for maintaining body temperature, metabolic activity, supporting growth, for physical work, to maintain constant body weight and good health.

9.7.1 Energy yielding food factors

The energy yielding food factors are (i) carbohydrates (ii) fats and(iii) proteins. Within the body, these are oxidised in the cells. The process is one of continuous utilization of O_2 and production of CO_2 H_2O and heat.





9.7.2 Units of energy – calorie and joule

The energy value of foods can be expressed in terms of kilocalories(KCal) or megajoules(MJ).The International Union of Nutritional Sciences has suggested the use of megajoule as the energy unit in place of Kcal. These units are defined as follows:

Kilocalorie: One kilogram calorie is the quantity of heat required to raise the temperature of 1 kg of water through 1° C.

Joule: A joule is defined as the energy required to move 1kg mass by 1 metre by a force of 1 Newton acting on it.

Newton: One Newton is the force needed to accelerate 1 kg mass by less than a second.

1 Kcal = 4.184 KJ

1000Kcal = 4.184 Megajoule(MJ)

1 MJ = 240Kcal

9.7.3 Energy value of foods

The energy in various foods is measured by calorimetry. Calorimetry is the measurement of heat loss.The energy value of foods is determined using the instrument called Bomb calorimeter.

9.7.4 Gross Energy value of foods

When samples of carbohydrate, fat, protein are burned, the amount of heat produced is always the same for each of these nutrients. The average gross energy value of carbohydrates,fats and proteins determined with bomb calorimeter is as follows:

1g of Carbohydrate = 4.1 kcal

1g of fat = 9.45 kcal

1g of protein = 5.65 kcal

9.7.5 Physiological energy value of foods

In the utilization of carbohydrates, fats and proteins in the body a certain percentage of these nutrients is lost in digestion and the nitrogen of protein is excreted in urine as urea which still contains some energy value.The average losses in digestion in human subjects have been estimated to be 2.0% for carbohydrates,5.0% for fats and 8.0% for proteins. The loss of energy in urea is estimated to be 1.2kcal per gram of protein oxidised. The physiological energy values of foods calculated from the gross energy values after allowing for the losses in digestion and metabolism are as follows: Carbohydrates 4.0;fats 9.0 and proteins 4.0.

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